Paper III – Subspeciality of Urology (practice paper, MCh Urology Summer 2017 set)

Practice paper compiled from the question bank: the actual 2017 Paper III was not available.

Contents

1. III-1. Classification of urethral strictures; surgical management after failed VIU (LAQ)
2. III-2. Female continence mechanism; classification of SUI; principles of surgical options (LAQ)
3. III-3a. Bladder myoplasty
4. III-3b. Pathogenesis and management of uretero-vaginal fistula
5. III-3c. Differential diagnosis and treatment of frequency and urgency in a 3-year-old boy
6. III-3d. Genetic causes of male infertility
7. III-3e. Urinary bladder compliance
8. III-3f. Congenital ureteral anomalies

III-1. Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU

Type: LAQ (25 marks)

Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar with variable **spongiofibrosis**.
- **VIU (visual internal urethrotomy)**, also called DVIU (direct vision internal urethrotomy), cuts the scar and lets it heal by **secondary intention**: a race between epithelialisation and wound contraction. It fails when contraction wins, which is usual when spongiofibrosis is deep.

Classification

1. By site (WHO (World Health Organization) 2004 / ICUD (International Consultation on Urological Diseases) 2010 terminology)

Segment	Term
Anterior: meatus, fossa navicularis, penile (pendulous), bulbar	**Stricture**
Membranous urethra after pelvic fracture	**PFUI (pelvic fracture urethral injury)** / distraction defect
Posterior urethra after prostatectomy or radiotherapy; bladder neck	**Stenosis or contracture**

2. By aetiology

Category	Examples
Idiopathic (commonest, with iatrogenic)	No cause found
Iatrogenic	Catheters, TURP (transurethral resection of prostate), cystoscopy, hypospadias repair, radiotherapy
Traumatic	Straddle injury (bulbar); pelvic fracture (PFUI)
Inflammatory	**Lichen sclerosus (LS)**; gonorrhoea (now rare)
Congenital	Rarest; infants only

3. By length: short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

4. By depth (Jordan): (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness spongiofibrosis, (E) fibrosis beyond the spongiosum, (F) complex stricture with fistula or abscess.

5. By complexity: simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

Why repeat VIU is futile after failure

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot

remove.

- **A second urethrotomy adds virtually nothing.** Each dilation or urethrotomy lengthens the scar and lowers later urethroplasty success. The "reconstructive ladder" is abandoned.
- AUA (American Urological Association) guideline (2016): offer **urethroplasty, not repeated endoscopy**, after failed dilation or DVIU.

Pre-operative evaluation

- Define the **4 Ls: location, length, depth, density**.
- **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)**, flexible urethroscopy, sonourethrography; MRI (magnetic resonance imaging) for PFUI.
- **Urethral rest**: suprapubic catheter for 6–8 weeks lets the true length show. Avoid instrumentation before surgery.
- Treat infection; look for LS at the meatus; examine the mouth; record erectile function.

Surgical options after failed VIU

Stricture	Operation	Principle
Bulbar, short (1–2 cm, up to 3–5 cm proximally)	**EPA (excision and primary anastomosis)**: gold standard, about 90–95% success	Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries
Bulbar, longer	**BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**); Kulkarni one-sided	Graft spread-fixed on a well-vascular bed (corpora or spongiosum)
Bulbar with an obliterated segment	**Augmented anastomosis**	Excise the narrowest part, anastomose the floor, graft the roof
Penile	One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular)	Avoid EPA here (chordee)
LS, failed hypospadias, infection	**Staged repair** (Johanson): graft, then tubularise after about 6 months	Oral mucosa, not genital skin, in LS
Panurethral	Kulkarni one-stage dorsal graft, or staged	—
Elderly, unfit, multiple failures	**Perineal urethrostomy** (BMG-augmented)	Simple, durable bypass
PFUI after failed VIU	Delayed **perineal anastomotic urethroplasty**	Bulbar mobilisation, then intracrural separation, inferior pubectomy, rerouting

- **Onlay is better than tube.** Graft and flap results are equal; grafts are simpler.
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon (**ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after repeated failed endoscopy); intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

Complications

- Recurrence, chordee, ED (erectile dysfunction; similar to circumcision overall), post-void dribble, graft-site pain and numbness.

Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR (post-void residual) and symptom score; urethroscopy if flow falls. EPA recurrences occur mostly in the first 6 months; graft results decline with time.

Recent advances

- Tissue-engineered oral mucosa; robotic posterior repair.

Diagram to draw: dorsal versus ventral onlay cross-section; EPA with spatulated ends.

> **Key points to remember:**
>
> - VIU cures only short bulbar strictures at first attempt; do not repeat it.
> - EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
> - Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
> - ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

III-2. Discuss continence mechanism in females . classify SUI and principles of different surgical options

Type: LAQ (25 marks)

Introduction

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, coughing or sneezing.
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.

A. Continence mechanism in the female

1. Intrinsic urethral factors

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-sensitive submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

2. Support (extrinsic)

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP (arcus tendineus fasciae pelvis)** and levator ani.
- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

3. Neural

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

4. Theories of continence and SUI

Theory	Mechanism	Surgical implication
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the mid-urethra	Tension-free MUS (mid-urethral sling)
ISD (intrinsic sphincter deficiency; McGuire & Lytton 1978)	Urethra itself fails to close	Sling, bulking, sphincter

- Most SUI has some ISD. **Hypermobility alone never causes SUI.**

B. Classification of SUI

1. Blaivas & Olsson (1988), on video-urodynamics

Type	Bladder neck at rest	On stress
0	Closed, above lower border of symphysis	Descends and opens, but no leak seen (history of SUI)

Type	Bladder neck at rest	On stress
I	Closed, above symphysis	Descends **<2 cm**; leaks; little or no cystocele
IIA	Closed, above symphysis	Descends **>2 cm** with cystourethrocele; leaks
IIB	Closed but **at or below** lower border of symphysis	Proximal urethra opens; leaks with or without further descent
III	**Open at rest** without detrusor contraction	Leaks with minimal stress = **ISD**

2. Functional: hypermobility versus ISD

	Urethral hypermobility	ISD
Q-tip test	Angle change **>30°**	Often fixed urethra
VLPP (Valsalva leak point pressure)	**≥90 cmH_2O**	**<60 cmH_2O** (60–90 equivocal)
MUCP (maximum urethral closure pressure)	Normal	**≤20 cmH_2O**
Cause	Childbirth, ageing, obesity	Previous surgery, radiation, neuropathy, ageing

3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR (post-void residual), radiation); pure versus mixed incontinence.

C. Principles of surgical options

Operation	How it works	Best for	Key facts
Burch colposuspension (open, laparoscopic or robotic)	Paravaginal fascia beside the bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing	Hypermobility; women declining mesh; at abdominal surgery	85–90% at 1 year, about 70% at 5 years; poor in ISD
MMK (Marshall–Marchetti–Krantz)	Bladder neck to pubic periosteum	Historical	Osteitis pubis; abandoned
Autologous pubovaginal sling (rectus fascia)	At the **bladder neck**, minimal tension; restores the hammock and compresses	**All types, including ISD**, recurrent SUI, neurogenic SUI	Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction
MUS: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (Delorme 2001), single-incision	Loose tape under the **mid-urethra** acts as a new pubourethral ligament; urethra kinks against it on stress	First-line index SUI	**TOMUS**: retropubic 80.8% vs transobturator 77.7% (equal); retropubic often preferred in ISD
Bulking agents	Submucosal injection at bladder neck or proximal urethra improves coaptation	ISD with fixed urethra; frail; minimal surgery	Low durability; repeat injections
AUS (artificial urinary sphincter)	Cuff round the bladder neck; constant circumferential closure	Severe recurrent ISD, fixed urethra	Second-line; erosion, infection, revision

- **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; mesh complications (exposure, erosion into bladder or urethra, pain) need excision; FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.
- Before surgery: PFMT (pelvic floor muscle training), weight loss, pessary.

Diagram to draw: DeLancey hammock cross-section; sites of Burch sutures, bladder-neck sling and mid-urethral tape.

> **Key points to remember:**
> - Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
> - Blaivas types 0, I, IIA, IIB, III; type III = ISD (VLPP <60, MUCP ≤20 cmH_2O).
> - Burch corrects position; slings and AUS correct function.
> - SISTEr (sling > Burch); TOMUS (retropubic = transobturator).

III-3a. Bladder myeloplsty

Type: SAQ (10 marks)

Assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).

Definition

- **Detrusor myoplasty**: wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

Indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap**: artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- Patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty**: simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

Other "myoplasty" procedure: detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

Diagram to draw: latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

Key points to remember:

- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

III-3b. Pathogenesis and management of uretero-vaginal fistula

Type: SAQ (10 marks)

Definition

- **UVF (uretero-vaginal fistula)**: abnormal communication between the ureter and vagina, causing continuous leakage.

Pathogenesis

- **Unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow abdominal hysterectomy**, often uneventful (ureteric injury 0.2–2.4%).
- Others: obstetric, cancer, radiation, endometriosis, foreign body.
- **Sites of injury:** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most in the **distal 6 cm**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation**. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- **Delayed recognition raises fistula risk about fourfold.**

Clinical features

- **Continuous leakage with normal voiding**: the other kidney fills the bladder.
- Appears **within 4 weeks**, often with fever, flank pain or ileus.

	VVF (vesicovaginal fistula)	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Investigations

- **Double dye test**: oral phenazopyridine (orange urine), blue dye into the bladder, tampon, walk. **Orange = UVF**; blue = VVF; both = both.
- **CT (computed tomography) urography**: hydronephrosis, extravasation.
- **Cystoscopy and RGP (retrograde pyelogram)**, stent at the same sitting.
- **Exclude coexisting VVF** (up to 25% of VVF have UVF) and bilateral injury.

Management

- **Stent first**: retrograde double-J if a wire passes; otherwise **PCN (percutaneous nephrostomy)** and antegrade stent. **Stenting within 2 weeks** works best. Remove at **4–6 weeks** and re-image.
- **Surgery** if stenting fails: **ureteroneocystostomy** with **psoas hitch** or **Boari flap**; nephropexy or ileal ureter for long gaps; open, laparoscopic or robotic. Early repair equals delayed. Success **≥90%**.
- Follow-up ultrasound: stricture may appear up to 30 months later.

Prevention

- Know the danger sites; cystoscopy at hysterectomy; selective prophylactic stents.

Diagram to draw: ureter under the uterine artery; double dye test.

Key points to remember:

- Distal ureteric injury at hysterectomy; leak within 4 weeks, normal voiding.
- Orange tampon = UVF; exclude coexisting VVF.
- Stent early; reimplant with psoas hitch if it fails.

III-3c. DD of frequency and urgency in 3 yr male child and treatment

Type: SAQ (10 marks)

Introduction

- **Frequency**: ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency**: sudden, compelling need to void.
- At 3 years toilet training may be ongoing; decide first whether the pattern is abnormal.

Differential diagnosis

Group	Causes and clues
Infection	**UTI (urinary tract infection)**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis
Functional	**BBD (bladder and bowel dysfunction)**/OAB (overactive bladder); voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting)
Bowel	**Constipation**: rectal mass presses on bladder; commonest associated cause
Obstruction	**PUV (posterior urethral valves)**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis
Neurogenic	Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality)
Bladder irritants	Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma**
Polyuria	Diabetes mellitus, diabetes insipidus, chronic kidney disease
Behavioural	Anxiety, family stress, developmental disorders

Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Bladder and bowel diary**; uroflow if the child cooperates.
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.

Treatment (cause-directed)

- **UTI:** antibiotics per culture.
- **Constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.
- **Urotherapy:** timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **OAB persisting after urotherapy: oxybutynin** 0.3–0.6 mg/kg/day in 2–3 doses (maximum 15 mg/day); watch for constipation.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.

- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.

Diagram to draw: flowchart: urinalysis → ultrasound with residual → targeted tests.

> **Key points to remember:**
> - Always ask about bowels; treat constipation first.
> - Rule out UTI, PUV, meatal stenosis and occult dysraphism.
> - Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

III-3d. Genetic causes of male infertility. Enumerate and classify.

Type: SAQ (10 marks)

Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

Classification

Group	Condition	Key features
1. Chromosomal, numerical	**Klinefelter syndrome (47,XXY)**	**Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction)
	47,XYY; **46,XX male** (SRY gene translocated to X)	XX male: azoospermia, no sperm retrievable
2. Chromosomal, structural	**Robertsonian and reciprocal translocations**, inversions, ring Y	Oligozoospermia; recurrent miscarriage
3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))	**AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes)	Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion
4. Single-gene: excurrent ducts	**CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction	Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508)
5. Single-gene: hypothalamic–pituitary	**Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations	Low LH (luteinising hormone) and FSH; treat with hCG ± FSH
6. Androgen action	**Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency	High testosterone and LH, poor sperm
7. Sperm structure and motility	**Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm)	ICSI; macrocephalic sperm: ICSI not advised (aneuploidy)

Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.

- **Also know:** Noonan syndrome, myotonic dystrophy.

Diagram to draw: Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

III-3e. Urinary bladder compliance

Type: SAQ (10 marks)

Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/cmH_2O**.
- It reflects **low-pressure storage** during filling.

Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cmH_2O** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cmH_2O**.

Pitfalls

Pitfall	Effect	Fix
Fast filling	False low compliance	Slow or stop filling; pressure falls
Sustained low-amplitude detrusor contraction	Mimics low compliance	Stop filling: pressure returns to baseline
Pop-off: VUR (vesicoureteric reflux), diverticulum, leaking outlet	Falsely good compliance	Video-urodynamics; occlude outlet

- A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.**

Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cmH_2O** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cmH_2O is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.

- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

Diagram to draw: cystometrogram: normal flat versus steep rising filling curve.

> **Key points to remember:**
> - C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
> - McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
> - Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

III-3f. Congenital ureteral anomalies

Type: SAQ (10 marks)

Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

Classification

Category	Anomaly	Key points
Number	**Duplication**: incomplete (bifid, Y-ureter) or complete	About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele
Termination (orifice)	**Ectopic ureter**	Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome
Structure	**Ureterocele**	Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens)
	PUJ obstruction	Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children)
	Megaureter (≥7 mm)	Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment
	Primary VUR	Short submucosal tunnel, lateral orifice
Position	**Preureteral (retrocaval) vena cava**	Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not
	Retroiliac ureter; herniation	Rare

Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.

- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

Diagram to draw: complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
>
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

Rapid Revision Sheet – Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU (LAQ)

- VIU cures only short bulbar strictures at first attempt; do not repeat it.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

III-2. Discuss continence mechanism in females . classify SUI and principles of different surgical options (LAQ)

- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD (VLPP <60, MUCP ≤20 cmH_2O).
- Burch corrects position; slings and AUS correct function.
- SISTEr (sling > Burch); TOMUS (retropubic = transobturator).

III-3a. Bladder myeloplsty (SAQ)

- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

III-3b. Pathogenesis and management of uretero-vaginal fistula (SAQ)

- Distal ureteric injury at hysterectomy; leak within 4 weeks, normal voiding.
- Orange tampon = UVF; exclude coexisting VVF.
- Stent early; reimplant with psoas hitch if it fails.

III-3c. DD of frequency and urgency in 3 yr male child and treatment (SAQ)

- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

III-3d. Genetic causes of male infertility. Enumerate and classify. (SAQ)

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

III-3e. Urinary bladder compliance (SAQ)

- $C = \Delta V/\Delta Pdet$; <20 mL/cmH_2O suggests poor compliance.
- McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

III-3f. Congenital ureteral anomalies (SAQ)

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.